

Scarring and Compounding: The Dynamic Economic Costs of Recurring Climate Exposure

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Abstract

Whether a changing climate raises costs over the long run depends on what happens once a shock ends: whether the damage reverses, leaves a one-time scar, or compounds as shocks recur. Using a U.S. county panel from 2011 to 2023, I decompose each climate shock into onset, persistence, and exit; test whether entering and leaving a shock offset one another; and estimate how outcomes respond to accumulated shock-years, checking the results against a staggered event-study estimator. The dynamics are shock-specific. Drought debt scars: the two-year onset-exit asymmetry in the medical-debt share is $+0.019$ ($p < 0.001$), about 45 percent of a typical annual movement. This is a scar that fades by the fourth year after exit and partly reflects out-migration. Cold, on the other hand, compounds: counties at 10+ cumulative cold-years have employment roughly 5,500 lower than counties having 1-3 years of exposure, and the gap approaches 5,000 jobs a decade after onset in the event study. Heat saturates into a level difference, and chronic extreme drought is extremely rare. The horizon over which climate raises costs is itself hazard-specific, so single-year exposure measures understate the burden of recurring cold.

Glossary of Abbreviations

Abbreviation	Expansion
ACA	Affordable Care Act; here, the individual health-insurance marketplace it created
ACS	American Community Survey (US Census Bureau)
AQI	Air Quality Index (US Environmental Protection Agency)
ATT	Average treatment effect on the treated
BEA	Bureau of Economic Analysis
BKY	Benjamini–Krieger–Yekutieli sharpened false-discovery-rate procedure
CDC	Centers for Disease Control and Prevention
CDD	Cooling degree days; the heat-exposure measure
CMS	Centers for Medicare & Medicaid Services
CONUS	Contiguous United States
CPI	Consumer Price Index, used to convert nominal to real dollars
CS	Callaway–Sant’Anna staggered difference-in-differences estimator
DiD	Difference-in-differences
DOE	US Department of Energy
DRDID	Doubly robust difference-in-differences estimator
ED	Emergency department
ERS	Economic Research Service (US Department of Agriculture)
FE	Fixed effects
HDD	Heating degree days; the cold-exposure measure
HIX	Health Insurance Exchange; source of marketplace premium data
IRS SOI	Internal Revenue Service Statistics of Income, used for migration flows
ITT	Intention to treat
LEAD	Low-Income Energy Affordability Data (US Department of Energy)
LP	Local projection
NASHP	National Academy for State Health Policy, source of hospital financial data
NHE	National Health Expenditure Accounts
NOAA	National Oceanic and Atmospheric Administration
NPR	Net patient revenue
PDSI	Palmer Drought Severity Index; extreme drought is a value at or below -4
PMPM	Per member per month, the unit in which premiums are compared to costs
pp	Percentage points

Abbreviation	Expansion
PRISM	Parameter-elevation Regressions on Independent Slopes Model, the gridded climate product used for humidity
PUF	Public use file
RA	Rating area, the geographic unit at which marketplace premiums are set
RI	Randomization inference
SAHIE	Small Area Health Insurance Estimates (US Census Bureau)
SD	Standard deviation
SE	Standard error
SVI	Social Vulnerability Index (CDC/ATSDR)
USDA	US Department of Agriculture
WCB	Wild cluster bootstrap, with Webb weights

1 Introduction

Whether a changing climate raises health and economic costs over the long run depends less on any single bad year than on what happens after it. When the drought breaks or the cold season ends, the damage may reverse, it may leave a one-time scar, or as shocks recur it may compound.

A large literature measures the contemporaneous and short-lag effects of temperature and drought on output and health (Dell et al., 2014), with a smaller strand on the persistence of local shocks (Deryugina et al., 2018; Hsiang and Jina, 2014). Much less is known about adjustment: whether the same shock’s effects unwind on exit, and whether recurrence raises the marginal harm. This essay provides that evidence for heat, cold, and drought.

Using a U.S. county panel from 2011 to 2023, I decompose each climate shock into its onset, its persistence, and its exit; test whether entering and leaving a shock offset one another, which I take as the operational definition of reversibility; estimate how outcomes respond to the cumulative number of shock-years a county has accumulated; and check the dose results against a staggered event-study estimator. The estimands are built to distinguish four adjustment regimes: reversal, scarring, saturation, and compounding.

The dynamics are shock-specific. Drought debt scars: the two-year onset-exit asymmetry in the medical-debt share is +0.019 ($p < 0.001$), about 45 percent of a typical annual movement. This is a scar that fades by the fourth year after exit and partly reflects out-migration. Cold, on the other hand, compounds: counties at 10+ cumulative cold-years have employment roughly 5,500 lower than counties having 1-3 years of exposure, and the gap approaches 5,000 jobs a decade after onset in the event study. Heat saturates into a level difference, and chronic extreme drought is extremely rare. The horizon over which climate raises costs is itself hazard-specific, so single-year exposure measures understate the burden of recurring cold.

This essay turns the informal language of scarring and compounding into operational estimands of symmetry tests, dose contrasts, and event-study gaps, and applies it uniformly across hazards in one unified panel. Section 2 explains why persistence matters; Section 3 sets out the transition framework and estimands; Section 4 summarizes the data and exposure support. Sections 5 through 7 report the symmetry, cumulative-dose, and long-run results; Section 8 examines migration and composition; Section 9 interprets the results by hazard; Section 10 concludes.

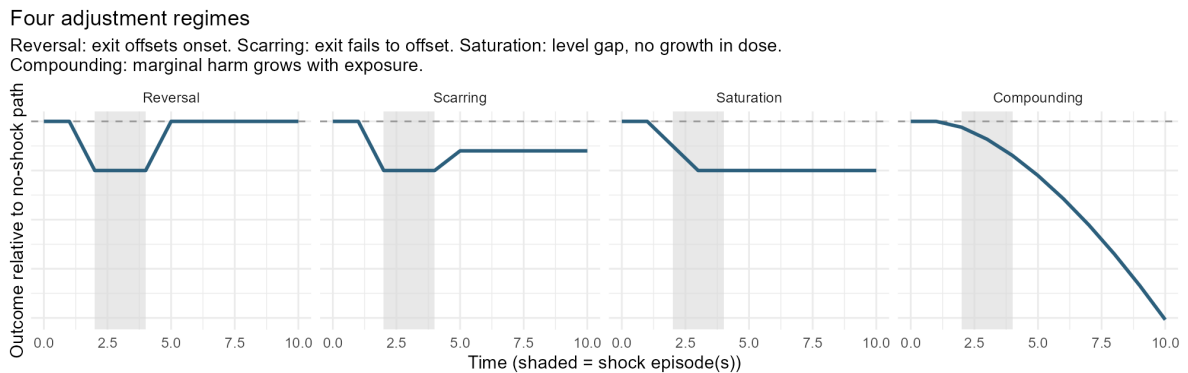
2 Why persistence matters

The welfare cost of a climate shock depends on what happens after it ends. A loss that reverses within a year and an equal-sized loss that persists have very different discounted burdens, and a shock whose marginal harm grows with recurrence is a different policy object from one whose costs arrive once. Further, if repeated exposure compounds, counties facing recurring hazards face rising marginal damages, and adaptation capital there pays a growing dividend. If effects saturate, the gap is a fixed level difference, and the policy question shifts toward transfers and insurance for the exposed geographies rather than protection against the next marginal year. Lastly, persistence also matters for measurement. Contemporaneous exposure metrics - this year's shock against this year's outcome - understate hazards whose harm accumulates across years. ([Dell et al., 2012](#))

3 Transition framework and estimands

I define four different types of adjustment that take place after a climate shock. Under "reversal", exit offsets onset and the shock leaves no lasting mark. Under "scarring", exit fails to offset onset and a gap remains after the shock ends. Under "saturation", effects persist at a level but do not grow with additional exposure. Under "compounding", the marginal harm grows as exposure recurs. Figure 1 illustrates the four paths.

Figure 1: Four adjustment paths after a shock: reversal, scarring, saturation, and compounding.



I decompose each hazard into onset, persistence, and exit indicators and test symmetry - whether the onset and exit coefficients sum to zero - as the operational definition of reversibility. A rejected symmetry test with a same-signed sum is scarring. Local projections then trace the post-exit path, which is what distinguishes a transient scar from a permanent one. ([Jordà, 2005](#))

$$Y_{it} = \alpha_i + \gamma_t + \beta^{\text{on}} O_{it} + \beta^{\text{pe}} R_{it} + \beta^{\text{ex}} E_{it} + \varepsilon_{it}, \quad H_0 : \beta^{\text{on}} + \beta^{\text{ex}} = 0 \quad (1)$$

Cumulative-dose estimands separate saturation from compounding: binned contrasts across accumulated shock-years and a smooth quadratic in dose. If effects saturate, high-dose and mid-dose counties look alike; if they compound, the gap grows with accumulated years. I interpret these as within-county exposure-history contrasts throughout.

$$Y_{it} = \alpha_i + \gamma_t + \sum_b \phi_b \mathbf{1}\{C_{it} \in b\} + \varepsilon_{it}, \quad C_{it} = \sum_{\tau \leq t} S_{i\tau} \quad (2)$$

Finally, staggered event-study estimates in the [Callaway and Sant’Anna \(2021\)](#) framework compare treated cohorts with not-yet-treated and never-treated counties over a decade of event time. These provide the long-run complement to the transition estimates, and emerge from an estimator built for staggered adoption.

4 Data and recurring exposure

The panel, shock definitions, and estimation conventions follow the first essay. Briefly: a county-year panel over 2011–2023 covering roughly 3,100 counties; four binary shock bins - heat and cold marked by national 80th-percentile degree-day cutoffs from a 1990–2000 pre-study baseline, extreme drought by an absolute Palmer threshold, and poor air quality by the EPA exceedance level; county and year fixed effects; and state-clustered standard errors. The first essay’s Appendix B shows the definitions are robust to longer baseline horizons and to the choice of estimation horizon.

Table 2 reports the transition support over 2011–2023. Drought is episodic in the data: 511 onset transitions and 705 exits against only 175 persisting county-years, across 603 counties ever exposed. Therefore, episodes begin and end quickly, and chronic extreme drought is rare: persisting spells are 0.5 percent of county-years, against 2.3 percent for any extreme-drought year. Heat and cold are different: 903 and 1,032 onsets sit alongside 8,125 and 5,485 persisting county-years, so exposure recurs and accumulates which is the variation the cumulative-dose estimates use. The transition regressions run on roughly 27,000 to 34,000 county-years, depending on outcome and horizon.

Table 2: Transition episodes by hazard, county panel 2011–2023

Hazard	Onset	Persist	Exit	Calm	County-years	Counties
Extreme drought	511	175	705	36,253	37,644	603
Extreme heat	903	8,125	1,033	27,583	37,644	1,016
Extreme cold	1,032	5,485	1,181	29,946	37,644	834

A county-year is classified by its position in an exposure episode: onset (first year of a spell), persistence (a continuing spell), exit (the first year after a spell ends), or calm. The counts show how much support each transition has, which is what limits the horizons the dynamic estimates can reach: drought spells are rare and short, while heat spells persist.

5 Onset–exit symmetry results

The symmetry test asks whether leaving a shock undoes what entering it did. For heat and cold employment the onset and exit coefficients offset; for drought and the credit-bureau debt share they do not. At a two-year horizon their sum - the asymmetry - is +0.019 ($p < 0.001$; Table 3), so the recorded debt share does not return to its pre-shock path within the horizon the test reaches. What follows is a statement about a ledger rather than a welfare measure: bureau debt records hardship only after an insurance relationship, a billed encounter, and a credit file, and the third essay shows the recorded response is weakest where coverage is thinnest. Part of the persistence is compositional, since drought raises net outflows the following year, so the estimate bounds same-population persistence rather than measuring it. The asymmetry survives a sharpened multiple-testing correction at $q < 0.05$, remains significant under no, lagged, and contemporaneous control sets estimated on the identical sample ($p < 0.001$ in each), though the point estimate moves between 0.019 and 0.025, and 99 percent of the lag-2 drought-debt coefficient survives premium adjustment.

The scar is large relative to how much the series normally moves. Scaled by the mean absolute deviation of year-to-year changes in the county’s debt share, the two-year asymmetry equals about 72 percent of a typical annual movement. It is also approximately 10 percent of the mean debt share of ~ 19 percent. Extending the local-projection horizon shows the drought impulse fading by the fourth year. At the three-year horizon, the furthest the symmetry test reaches, the asymmetry remains significant (+0.015, $p = 0.039$). Table 3 reports symmetry tests for the full grid of hazards and outcomes.

Table 3: Onset–exit symmetry tests at a two-year horizon

Hazard	Outcome	Weighting	Onset	Exit	Asymmetry	p	Rejected
Extreme drought	Civilian employment	Population	2,392	23,022	25,414	0.382	no
Extreme drought	Civilian employment	Unweighted	157	713	870	0.377	no
Extreme drought	Medical debt share	Population	0.00796	0.00542	0.0134	<0.001	yes
Extreme drought	Medical debt share	Unweighted	0.0137	0.00503	0.0187	<0.001	yes
Extreme drought	Per-capita income	Population	293	971	1,264	0.248	no
Extreme drought	Per-capita income	Unweighted	476	1,147	1,623	0.002	yes
Extreme heat	Civilian employment	Population	-1,327	805	-522	0.912	no
Extreme cold	Civilian employment	Population	4,758	1,103	5,861	0.382	no
Extreme heat	Civilian employment	Unweighted	1,010	1,407	2,417	0.006	yes
Extreme cold	Civilian employment	Unweighted	192	-158	34.4	0.955	no
Extreme heat	Medical debt share	Population	0.00364	0.00873	0.0124	0.088	no
Extreme cold	Medical debt share	Population	-0.00127	0.000874	-0.000398	0.964	no
Extreme heat	Medical debt share	Unweighted	0.00304	0.00759	0.0106	0.048	yes
Extreme cold	Medical debt share	Unweighted	-0.00829	-0.00336	-0.0117	0.118	no
Extreme heat	Per-capita income	Population	131	481	612	0.289	no
Extreme cold	Per-capita income	Population	151	60.4	212	0.717	no
Extreme heat	Per-capita income	Unweighted	256	193	448	0.114	no
Extreme cold	Per-capita income	Unweighted	651	396	1,047	0.005	yes

Symmetry requires that the effect of entering a shock state and the effect of leaving it offset one another, so that their sum is zero. The asymmetry column reports that sum; a value reliably different from zero means the outcome does not return to its pre-shock path when the shock ends. Estimates at other horizons and for the remaining outcomes are reported in the analysis output.

6 Cumulative-dose results

Cumulative dose asks whether the marginal cold year hurts more as a county’s exposure history accumulates - a different question from whether a single episode unwinds. I count the cold-shock years each county has accumulated over the panel and compare outcomes across accumulated doses, using a binned contrast - ten or more cumulative cold-years versus one to three - and a smooth quadratic dose term, with county and year fixed effects throughout.

In the unweighted binned contrast, a county at ten or more cumulative cold-years has employment 5,522 lower than a county at one to three (SE 1,196, $p = <0.01$) - about 11 percent of mean county employment of 48,068. The population-weighted contrast is far larger ($-41,573$, $p = 0.005$), reflecting the weight of large counties in levels-based outcomes. The smooth quadratic dose term implies a marginal-effect difference between the tenth and first cold-year of $+417$ - flat and statistically insignificant. The binned estimator survives humidity adjustment, an adjustment for population aging, the stricter 90th-percentile definition of extreme cold, and all three control variants estimated on the identical sample ($p < 0.001$ in each).

7 Long-run and recurring-treatment comparisons

A staggered difference-in-differences in the [Callaway and Sant’Anna \(2021\)](#) framework traces the employment gap in event time. Employment in cold-treated counties is 4,982 lower a decade after onset ($p = 0.003$), and the gap compounds with time from roughly 150 in the onset year to 2,600 at five years and 4,900 at ten. The decade-out estimate rests on a single pre-year by construction; pooling the available pre-period leaves it essentially unchanged ($-4,990$, $p = 0.005$). The medical-debt share is also higher a decade after onset, but that gap is flat through 2022 and appears entirely in 2023, the year credit-bureau medical-debt reporting rules changed, so I do not read it as compounding evidence and note it only with that caveat.

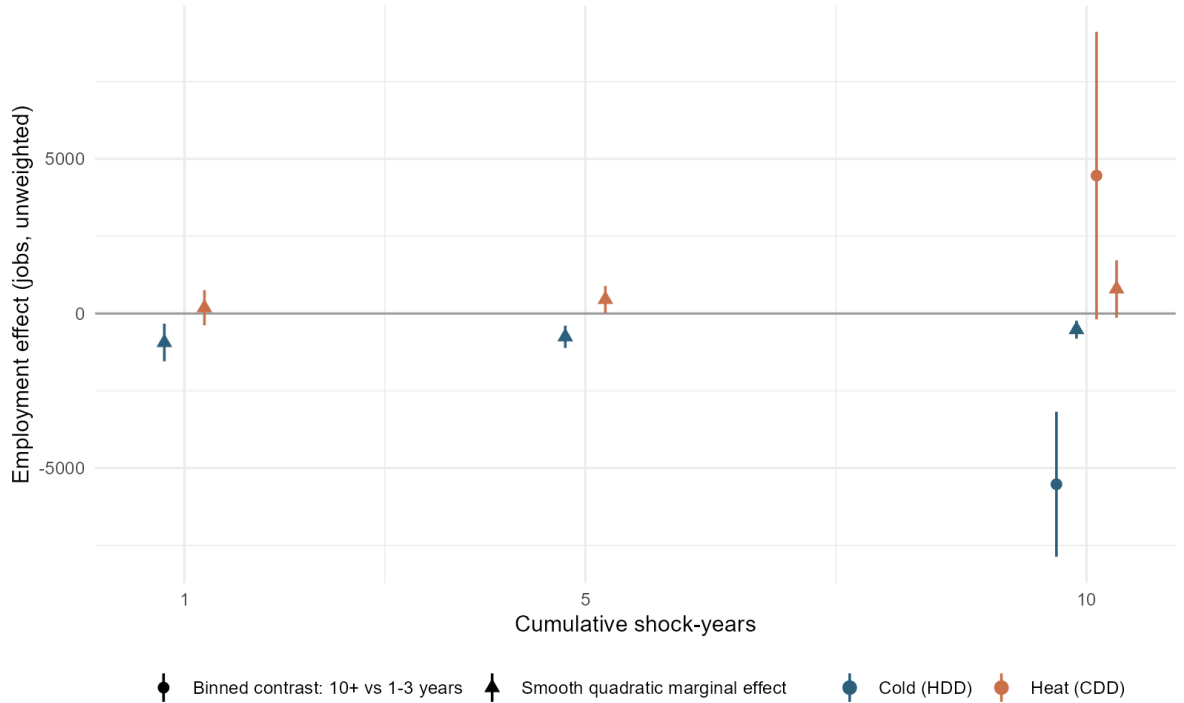
Horizon choices do not drive these verdicts. Extending estimation horizons change the short-horizon estimates by less than one standard error, and cold employment losses persist at three- and four-year horizons in local projections ($p = 0.009$ and 0.006). Only shortening matters: truncating the horizon at two years understates cold compounding.

Heat behaves differently. Its cumulative-dose profile shows no negative gradient - the binned ten-plus versus one-to-three contrast is +4,460 jobs ($p = 0.06$), against cold's $-5,522$ (Figure 2) - so heat's persistent gap is a level difference rather than one that grows with accumulated exposure (saturation). Chronic extreme drought, by contrast, is too rare in the panel to estimate a dose response at all - only 175 persisting county-years against 511 onsets - so drought in these data is episodic, which is why its dynamics were examined through onset and exit rather than accumulation.

Figure 2: Cumulative-exposure contrasts for cold and heat: employment falls with accumulated cold-years but not with accumulated heat-years.

Cold compounds with cumulative exposure; heat does not

Within-county exposure-history contrasts (county+year FE). Binned 10+ vs 1-3 years: cold $-5,522$ jobs ($p=3.9e-06$); heat $4,460$ jobs ($p=0.06$) - no negative dose gradient. Smooth-quadratic MEs flat for both.



8 Migration, composition, and survivorship

Some of the persistence has a demographic component. IRS county-to-county migration flows show drought lowering the net in-migration rate by 0.21 percentage points the following year ($p = 0.047$) - that is, raising net outflows. I read this as a bounded, suggestive contributor to the drought scar rather than a decomposition of it. (Mullins and Bharadwaj, 2021) Part of the persistent debt gap may therefore be composition as households that remain differ in characteristics from those who left. The same logic applies to the long-run cold gaps, for two further reasons. First, the compounding

estimate is not invariant to how counties are weighted, as Section 7 sets out. Second, the Callaway-Sant’Anna aggregation does not hold the treated population fixed as event time lengthens: the admissible cold-onset cohorts are 2013 (171 counties) and 2014 (59 counties), and because the panel ends in 2023, only the 2013 cohort has ten years of post-onset data to contribute an event-time-10 estimate — a county onsetting in 2014 drops out of that average by construction.

9 Interpretation by hazard

The dynamics are hazard-specific. Drought is episodic: its recorded-debt scar is real at a two-year horizon, fades by the fourth year, and is partly compositional. Cold compounds under the binned cumulative-dose contrast and the staggered event-study estimator with employment gaps that widen a decade after onset. Heat saturates into a level difference that does not grow with additional exposure.

It follows that the policy-relevant horizon is itself hazard-specific. Single-year exposure measures understate the burden of recurring cold, while long averages dilute the episodic sharpness of drought. Adaptation targeting inherits the same logic: compounding favors protecting counties with recurring exposure before the next marginal year arrives, while saturation favors supporting the already-exposed stock.

10 Conclusion

This essay asked what happens when a climate shock ends or recurs: whether its economic effects reverse, scar, saturate, or compound. The answers are hazard-specific. Drought debt scars - a two-year onset-exit asymmetry of $+0.019$, about 45 percent of a typical annual movement in the debt share, fading by the fourth year after exit. Cold employment compounds, with a binned cumulative-dose gap of roughly 5,500 jobs and an event-study gap of roughly 5,000 a decade after onset. Heat saturates into a level difference.

The broader implication is about measurement. Because the horizon over which climate raises costs is hazard-specific, exposure measures that stop at the current year systematically understate recurring hazards. Longer panels and working-age administrative data would extend the horizons

these tests can reach.

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